HAT-P-36b

R-1159

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_220516` · created 2026-07-31T02:06:52+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459715.7921 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1571 +/- 0.0097
Transit depth (Rp/Rs)^2	2.47 +/- 0.31 [%]
Orbital Inclination (inc)	89.83 +/- 2.41
Airmass coefficient 1 (a1)	0.967 +/- 0.011
Airmass coefficient 2 (a2)	0.0206 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#3 - [304, 404]
Optimal Method	PSF photometry
Transit Duration (day)	0.0431 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1571 ± 0.0097 vs 0.126 (deviation 3.21σ)
Duration fit vs published	0.0431 d vs 0.0925 d (deviation 13.0σ)
Mid-time O–C	-4.8 min
Depth SNR	16.2
Residual scatter	1.11 %

Light curve

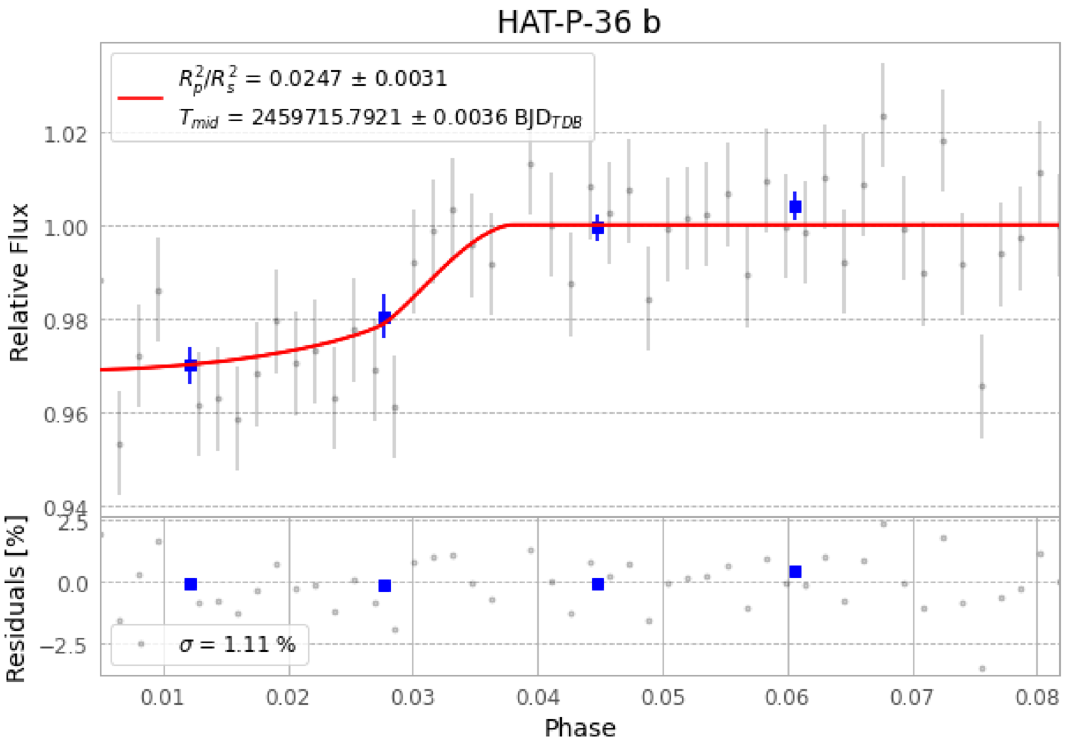

Figure 1 — FinalLightCurve_HAT-P-36 b_2022-05-16.png (SHA-256 4620ac70ab36b204…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [447, 258], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 13d90d1d30ffc2d3…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

50 FITS frames (33 MB), `HATP-36220516070612.FITS` → `HATP-36220516093312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-220516.log (d9c0435222ae…), FinalLightCurve_HAT-P-36 b_2022-05-16.png (4620ac70ab36…), FinalLightCurve_HAT-P-36 b_2022-05-16.csv (0621f89cc03b…)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)